

Lecture 11: conjugate gradient methods for multi-variable unconstrained optimization problem

MA3236 Non-Linear Programming

Hu Yukuan

yukuan.hu@nus.edu.sg

S17-08-21

Quick review

Steepest descent (SD) chooses

$$\mathbf{d}^{(k)} = -\mathbf{g}^{(k)}, \quad \mathbf{g}^{(k)} = \nabla f(\mathbf{x}^{(k)}).$$

- $\mathbf{H}_f \Rightarrow \mathbf{I}$ or fastest first-order decrease.
- Armijo backtracking on L -smooth functions.

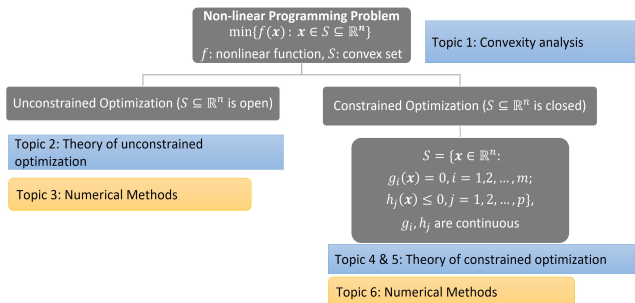
$$f\left(\mathbf{x} - \frac{1}{L}\nabla f(\mathbf{x})\right) \leq f(\mathbf{x}) - \frac{1}{2L}\|\nabla f(\mathbf{x})\|^2.$$

- μ -strong convexity + L -smoothness gives

$$E^{(k)} \leq E^{(k)} - \frac{1}{2L}\|\mathbf{g}^{(k)}\|^2, \quad \|\mathbf{g}^{(k)}\|^2 \geq 2\mu E^{(k)} \rightsquigarrow E^{(k)} \leq \left(1 - \frac{\mu}{L}\right)^k E^{(0)}.$$

$\Rightarrow$ preconditioned SD.

Topics



In Topic 3, we will study numerical methods for solving **unconstrained optimization problem**.

- Methods for solving one-variable unconstrained optimization.
- Methods for solving multi-variable unconstrained optimization: Newton method, steepest descent method, **conjugate-gradient method**

Newton and SD

Newton

- gradient and Hessian
- Hessian evaluation and a linear solve
- locally **quadratic** under regularity
- sensitive to regularity and the starting point

SD

- **gradient only**
- gradient evaluation and line search
- **globally** linear, or possibly sublinear
- sensitive to landscape conditioning

Can an algorithm retain **inexpensive iterations** while **favoring fast (global) convergence**?

Today's main scope — quadratic minimization

Let

$$q(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} - \mathbf{c}^T \mathbf{x}, \quad \mathbf{Q} = \mathbf{Q}^T \succ 0.$$

Then

$$\mathbf{g}(\mathbf{x}) = \nabla q(\mathbf{x}) = \mathbf{Q} \mathbf{x} - \mathbf{c}, \quad \nabla q(\mathbf{x}^*) = \mathbf{0} \iff \mathbf{Q} \mathbf{x}^* = \mathbf{c}.$$

Positive definiteness gives a unique minimizer, positive curvature along every nonzero direction, and a valid $\mathbf{Q}$ -geometry.

A perfect playground of both numerical linear algebra and optimization.

Memory through expanding subspaces

SD optimizes one line at a time.

$$\mathbf{x}^{(k)} = \arg \min_{\mathbf{x} \in \mathbf{x}^{(k-1)} + \text{span}\{\mathbf{d}^{(k-1)}\}} q(\mathbf{x}).$$

Conjugate directions ask for more:

$$\mathbf{x}^{(k)} = \arg \min_{\mathbf{x} \in \mathbf{x}^{(0)} + \text{span}\{\mathbf{d}^{(0)}, \dots, \mathbf{d}^{(k-1)}\}} q(\mathbf{x}).$$

Thus each iterate remains optimal over every direction used so far.

First-order form

At the subspace minimizer,

$$(\mathbf{g}^{(k)})^T \mathbf{d}^{(i)} = 0, \quad i = 0, \dots, k-1.$$

The first line search

Start from $\mathbf{x}^{(0)}$ and choose a nonzero direction $\mathbf{d}^{(0)}$. Set

$$\alpha^{(0)} = \arg \min_{\alpha \in \mathbf{R}} q(\mathbf{x}^{(0)} + \alpha \mathbf{d}^{(0)}), \quad \mathbf{x}^{(1)} = \mathbf{x}^{(0)} + \alpha^{(0)} \mathbf{d}^{(0)}.$$

Exact line search gives

$$(\mathbf{g}^{(1)})^T \mathbf{d}^{(0)} = 0.$$

Now choose $\mathbf{d}^{(1)}$ and perform a second exact line search.

The first line search

Start from $\mathbf{x}^{(0)}$ and choose a nonzero direction $\mathbf{d}^{(0)}$. Set

$$\alpha^{(0)} = \arg \min_{\alpha \in \mathbf{R}} q(\mathbf{x}^{(0)} + \alpha \mathbf{d}^{(0)}), \quad \mathbf{x}^{(1)} = \mathbf{x}^{(0)} + \alpha^{(0)} \mathbf{d}^{(0)}.$$

Exact line search gives

$$(\mathbf{g}^{(1)})^T \mathbf{d}^{(0)} = 0.$$

Now choose $\mathbf{d}^{(1)}$ and perform a second exact line search.

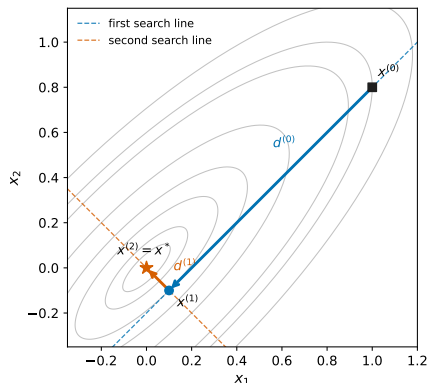
For $\mathbf{d}^{(k)} \neq \mathbf{0}$, minimize

$$\phi^{(k)}(\alpha) = q(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)}) = q(\mathbf{x}^{(k)}) + \alpha (\mathbf{g}^{(k)})^T \mathbf{d}^{(k)} + \frac{\alpha^2}{2} (\mathbf{d}^{(k)})^T \mathbf{Q} \mathbf{d}^{(k)}, \quad \alpha \in \mathbf{R}.$$

Since $\mathbf{Q} \succ 0$, the unique minimizer is

$$\alpha^{(k)} = - \frac{(\mathbf{g}^{(k)})^T \mathbf{d}^{(k)}}{(\mathbf{d}^{(k)})^T \mathbf{Q} \mathbf{d}^{(k)}}.$$

The two-direction memory requirement



We want $\mathbf{x}^{(2)}$ to minimize q over $\mathbf{x}^{(0)} + \text{span}\{\mathbf{d}^{(0)}, \mathbf{d}^{(1)}\}$. Hence

$$(\mathbf{g}^{(2)})^T \mathbf{d}^{(0)} = 0, \quad (\mathbf{g}^{(2)})^T \mathbf{d}^{(1)} = 0.$$

The second equality comes directly from the [second exact line search](#).

The conjugacy condition

Gradient update

$$\mathbf{g}^{(k+1)} = \mathbf{Q}\mathbf{x}^{(k+1)} - \mathbf{c} = \mathbf{Q}(\mathbf{x}^{(k)} + \alpha^{(k)}\mathbf{d}^{(k)}) - \mathbf{c} = \mathbf{g}^{(k)} + \alpha^{(k)}\mathbf{Q}\mathbf{d}^{(k)}.$$

The quadratic gradient update and the first exact line search give

$$0 = (\mathbf{g}^{(2)})^T \mathbf{d}^{(0)} = \left(\mathbf{g}^{(1)} + \alpha^{(1)}\mathbf{Q}\mathbf{d}^{(1)} \right)^T \mathbf{d}^{(0)}$$

The conjugacy condition

Gradient update

$$\mathbf{g}^{(k+1)} = \mathbf{Q}\mathbf{x}^{(k+1)} - \mathbf{c} = \mathbf{Q}(\mathbf{x}^{(k)} + \alpha^{(k)}\mathbf{d}^{(k)}) - \mathbf{c} = \mathbf{g}^{(k)} + \alpha^{(k)}\mathbf{Q}\mathbf{d}^{(k)}.$$

The quadratic gradient update and the first exact line search give

$$\begin{aligned} 0 = (\mathbf{g}^{(2)})^T \mathbf{d}^{(0)} &= \left(\mathbf{g}^{(1)} + \alpha^{(1)} \mathbf{Q} \mathbf{d}^{(1)} \right)^T \mathbf{d}^{(0)} \\ &= \underbrace{(\mathbf{g}^{(1)})^T \mathbf{d}^{(0)}}_{=0} + \alpha^{(1)} (\mathbf{d}^{(1)})^T \mathbf{Q} \mathbf{d}^{(0)}. \end{aligned}$$

The conjugacy condition

Gradient update

$$\mathbf{g}^{(k+1)} = \mathbf{Q}\mathbf{x}^{(k+1)} - \mathbf{c} = \mathbf{Q}(\mathbf{x}^{(k)} + \alpha^{(k)}\mathbf{d}^{(k)}) - \mathbf{c} = \mathbf{g}^{(k)} + \alpha^{(k)}\mathbf{Q}\mathbf{d}^{(k)}.$$

The quadratic gradient update and the first exact line search give

$$\begin{aligned} 0 &= (\mathbf{g}^{(2)})^T \mathbf{d}^{(0)} = \left(\mathbf{g}^{(1)} + \alpha^{(1)}\mathbf{Q}\mathbf{d}^{(1)} \right)^T \mathbf{d}^{(0)} \\ &= \underbrace{(\mathbf{g}^{(1)})^T \mathbf{d}^{(0)}}_{=0} + \alpha^{(1)}(\mathbf{d}^{(1)})^T \mathbf{Q}\mathbf{d}^{(0)}. \end{aligned}$$

For a nonzero second step, $\alpha^{(1)} \neq 0$, memory therefore requires

$$(\mathbf{d}^{(1)})^T \mathbf{Q}\mathbf{d}^{(0)} = 0.$$

Imposing this condition guarantees that the new exact line search preserves the earlier one.

Conjugate directions and $\mathbf{Q}$ -geometry

For $\mathbf{Q} \succ 0$, define

$$\langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{Q}} := \mathbf{u}^T \mathbf{Q} \mathbf{v}, \quad \|\mathbf{u}\|_{\mathbf{Q}} := \sqrt{\mathbf{u}^T \mathbf{Q} \mathbf{u}}.$$

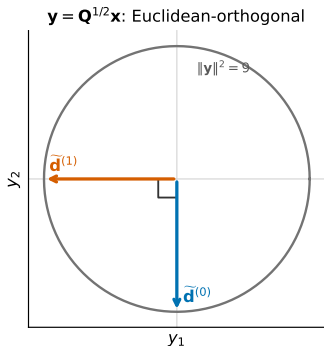
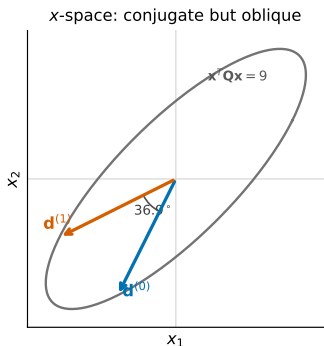
Nonzero directions $\mathbf{d}^{(0)}, \dots, \mathbf{d}^{(m)}$ are $\mathbf{Q}$ -conjugate if

$$(\mathbf{d}^{(i)})^T \mathbf{Q} \mathbf{d}^{(j)} = 0, \quad i \neq j.$$

They are orthogonal in the geometry of the quadratic level sets, although they need not be Euclidean-orthogonal.

Conjugate directions become orthogonal under $\mathbf{Q}^{1/2}$

$$\mathbf{Q} = \begin{pmatrix} 5 & -4 \\ -4 & 5 \end{pmatrix}, \quad \mathbf{Q}^{1/2} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad \mathbf{d}^{(0)} = \begin{pmatrix} -1 \\ -2 \end{pmatrix}, \quad \mathbf{d}^{(1)} = \begin{pmatrix} -2 \\ -1 \end{pmatrix}.$$



With $\tilde{\mathbf{d}}^{(i)} := \mathbf{Q}^{1/2} \mathbf{d}^{(i)}$,

$$(\tilde{\mathbf{d}}^{(0)})^T \tilde{\mathbf{d}}^{(1)} = (\mathbf{d}^{(0)})^T \mathbf{Q} \mathbf{d}^{(1)} = 0,$$

$$(\mathbf{d}^{(0)})^T \mathbf{d}^{(1)} = 4 \neq 0.$$

Conjugate directions' memory mechanism

Choose pairwise **Q-conjugate directions** $\mathbf{d}^{(0)}, \dots, \mathbf{d}^{(n-1)}$, and perform an **exact line search** along each direction in sequence.

- The current gradient is orthogonal to every past direction.

$$(\mathbf{g}^{(k)})^T \mathbf{d}^{(i)} = 0, \quad i = 0, \dots, k-1.$$

- The current iterate minimizes q over the affine subspace spanned by all directions used so far.

$$\mathbf{x}^{(k)} = \arg \min_{\mathbf{x} \in \mathbf{x}^{(0)} + \text{span}\{\mathbf{d}^{(0)}, \dots, \mathbf{d}^{(k-1)}\}} q(\mathbf{x}).$$

Conjugate-direction method

Given $\mathbf{x}^{(0)}$ and nonzero pairwise $\mathbf{Q}$ -conjugate directions $\mathbf{d}^{(0)}, \dots, \mathbf{d}^{(n-1)}$:

① Compute $\mathbf{g}^{(0)} = \mathbf{Q}\mathbf{x}^{(0)} - \mathbf{c}$.

② For $k = 0, 1, \dots, n - 1$:

$$\alpha^{(k)} = -\frac{(\mathbf{g}^{(k)})^T \mathbf{d}^{(k)}}{(\mathbf{d}^{(k)})^T \mathbf{Q} \mathbf{d}^{(k)}},$$

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha^{(k)} \mathbf{d}^{(k)}, \quad \mathbf{g}^{(k+1)} = \mathbf{g}^{(k)} + \alpha^{(k)} \mathbf{Q} \mathbf{d}^{(k)}.$$

③ Stop early if $\mathbf{g}^{(k+1)} = \mathbf{0}$.

Finite termination and finite precision

- Nonzero $\mathbf{Q}$ -conjugate directions are linearly independent.

$$\mathbf{0} = \sum_{i=0}^{n-1} c_i \mathbf{d}^{(i)} \Rightarrow \mathbf{0} = \sum_{i=0}^{n-1} c_i \mathbf{Q} \mathbf{d}^{(i)} \Rightarrow c_i = 0, \forall i.$$

- In exact arithmetic, n such directions span $\mathbf{R}^n$, so the method reaches $\mathbf{x}^*$ in at most n steps.

Finite termination and finite precision

- Nonzero $\mathbf{Q}$ -conjugate directions are linearly independent.

$$\mathbf{0} = \sum_{i=0}^{n-1} c_i \mathbf{d}^{(i)} \Rightarrow \mathbf{0} = \sum_{i=0}^{n-1} c_i \mathbf{Q} \mathbf{d}^{(i)} \Rightarrow c_i = 0, \forall i.$$

- In exact arithmetic, n such directions span $\mathbf{R}^n$, so the method reaches $\mathbf{x}^*$ in at most n steps.
- Floating-point errors gradually destroy orthogonality and conjugacy.
- Implementations stop using a relative gradient-residual tolerance.

A common quadratic example

Consider

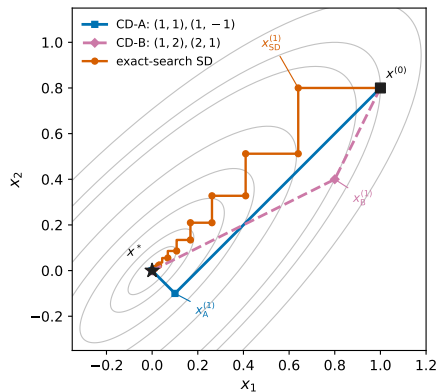
$$q(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x}, \quad \mathbf{Q} = \begin{pmatrix} 5 & -4 \\ -4 & 5 \end{pmatrix}, \quad \mathbf{x}^{(0)} = \begin{pmatrix} 1 \\ 0.8 \end{pmatrix}.$$

The eigenvalues are 1 and 9, so

$$\mathbf{x}^* = \mathbf{0}, \quad q(\mathbf{x}^{(0)}) = 0.9, \quad \kappa(\mathbf{Q}) = 9.$$

We use this example for conjugate directions (CD) and SD.

Different methods, different paths



method	$q(\mathbf{x}^{(1)})$	$q(\mathbf{x}^{(2)})$	exact result
CD-A	0.09	0	two steps
CD-B	0.72	0	two steps
exact-search SD	0.576	0.36864	continues

A wishlist for conjugate directions

We want directions that are

- **recursive**, using the current gradient and the last direction rather than the full history
- **inexpensive**, using matrix-vector products instead of a factorization

A naive choice: $(\hat{\mathbf{d}}^{(i)})^T \hat{\mathbf{d}}^{(j)} = \delta_{ij}, \quad \mathbf{d}^{(i)} := \mathbf{Q}^{-1/2} \hat{\mathbf{d}}^{(i)}.$

- **conjugate**, so earlier subspace minimizations remain valid

Conjugate gradient (CG) satisfies these requirements for PD quadratics!

The conjugate-gradient (CG) recurrence

Start from $\mathbf{d}^{(0)} = -\mathbf{g}^{(0)}$. For $k \geq 1$, combine the current negative gradient with the preceding direction:

$$\mathbf{d}^{(k)} = -\mathbf{g}^{(k)} + \beta^{(k)}\mathbf{d}^{(k-1)}.$$

The conjugate-gradient (CG) recurrence

Start from $\mathbf{d}^{(0)} = -\mathbf{g}^{(0)}$. For $k \geq 1$, combine the current negative gradient with the preceding direction:

$$\mathbf{d}^{(k)} = -\mathbf{g}^{(k)} + \beta^{(k)}\mathbf{d}^{(k-1)}.$$

Imposing adjacent conjugacy, $(\mathbf{d}^{(k)})^T \mathbf{Q} \mathbf{d}^{(k-1)} = 0$, first gives

$$\beta^{(k)} = \frac{(\mathbf{g}^{(k)})^T \mathbf{Q} \mathbf{d}^{(k-1)}}{(\mathbf{d}^{(k-1)})^T \mathbf{Q} \mathbf{d}^{(k-1)}}, \quad \alpha^{(k)} = -\frac{(\mathbf{g}^{(k)})^T \mathbf{d}^{(k)}}{(\mathbf{d}^{(k)})^T \mathbf{Q} \mathbf{d}^{(k)}}.$$

The conjugate-gradient (CG) recurrence

Start from $\mathbf{d}^{(0)} = -\mathbf{g}^{(0)}$. For $k \geq 1$, combine the current negative gradient with the preceding direction:

$$\mathbf{d}^{(k)} = -\mathbf{g}^{(k)} + \beta^{(k)}\mathbf{d}^{(k-1)}.$$

Imposing adjacent conjugacy, $(\mathbf{d}^{(k)})^T \mathbf{Q} \mathbf{d}^{(k-1)} = 0$, first gives

$$\beta^{(k)} = \frac{(\mathbf{g}^{(k)})^T \mathbf{Q} \mathbf{d}^{(k-1)}}{(\mathbf{d}^{(k-1)})^T \mathbf{Q} \mathbf{d}^{(k-1)}}, \quad \alpha^{(k)} = -\frac{(\mathbf{g}^{(k)})^T \mathbf{d}^{(k)}}{(\mathbf{d}^{(k)})^T \mathbf{Q} \mathbf{d}^{(k)}}.$$

$$\begin{aligned} \mathbf{g}^{(k+1)} &= \mathbf{g}^{(k)} + \alpha^{(k)} \mathbf{Q} \mathbf{d}^{(k)}, \\ (\mathbf{g}^{(k)})^T \mathbf{d}^{(i)} &= 0, \quad (\mathbf{g}^{(k)})^T \mathbf{g}^{(i)} = 0, \\ (\mathbf{d}^{(k)})^T \mathbf{Q} \mathbf{d}^{(i)} &= 0, \end{aligned}$$

residual recurrence,
orthogonality,
mutual conjugacy.

Why the step length simplifies

Exact line search along $\mathbf{d}^{(k)}$ gives

$$\alpha^{(k)} = -\frac{(\mathbf{g}^{(k)})^T \mathbf{d}^{(k)}}{(\mathbf{d}^{(k)})^T \mathbf{Q} \mathbf{d}^{(k)}}.$$

For $k \geq 1$, use the recurrence and $(\mathbf{g}^{(k)})^T \mathbf{d}^{(k-1)} = 0$:

$$-(\mathbf{g}^{(k)})^T \mathbf{d}^{(k)} = -(\mathbf{g}^{(k)})^T (-\mathbf{g}^{(k)} + \beta^{(k)} \mathbf{d}^{(k-1)})$$

Why the step length simplifies

Exact line search along $\mathbf{d}^{(k)}$ gives

$$\alpha^{(k)} = -\frac{(\mathbf{g}^{(k)})^T \mathbf{d}^{(k)}}{(\mathbf{d}^{(k)})^T \mathbf{Q} \mathbf{d}^{(k)}}.$$

For $k \geq 1$, use the recurrence and $(\mathbf{g}^{(k)})^T \mathbf{d}^{(k-1)} = 0$:

$$\begin{aligned} -(\mathbf{g}^{(k)})^T \mathbf{d}^{(k)} &= -(\mathbf{g}^{(k)})^T (-\mathbf{g}^{(k)} + \beta^{(k)} \mathbf{d}^{(k-1)}) \\ &= (\mathbf{g}^{(k)})^T \mathbf{g}^{(k)}. \end{aligned}$$

(The same identity is immediate for $k = 0$.) Hence

$$\alpha^{(k)} = \frac{(\mathbf{g}^{(k)})^T \mathbf{g}^{(k)}}{(\mathbf{d}^{(k)})^T \mathbf{Q} \mathbf{d}^{(k)}}.$$

Why the conjugate coefficient simplifies

Start from the adjacent-conjugacy coefficient, with the index shifted:

$$\beta^{(k)} = \frac{(\mathbf{g}^{(k)})^T \mathbf{Q} \mathbf{d}^{(k-1)}}{(\mathbf{d}^{(k-1)})^T \mathbf{Q} \mathbf{d}^{(k-1)}}.$$

The residual recurrence and $(\mathbf{g}^{(k)})^T \mathbf{g}^{(k-1)} = 0$ give

$$(\mathbf{g}^{(k)})^T \mathbf{Q} \mathbf{d}^{(k-1)} = \frac{(\mathbf{g}^{(k)})^T (\mathbf{g}^{(k)} - \mathbf{g}^{(k-1)})}{\alpha^{(k-1)}}$$

Why the conjugate coefficient simplifies

Start from the adjacent-conjugacy coefficient, with the index shifted:

$$\beta^{(k)} = \frac{(\mathbf{g}^{(k)})^T \mathbf{Q} \mathbf{d}^{(k-1)}}{(\mathbf{d}^{(k-1)})^T \mathbf{Q} \mathbf{d}^{(k-1)}}.$$

The residual recurrence and $(\mathbf{g}^{(k)})^T \mathbf{g}^{(k-1)} = 0$ give

$$(\mathbf{g}^{(k)})^T \mathbf{Q} \mathbf{d}^{(k-1)} = \frac{(\mathbf{g}^{(k)})^T (\mathbf{g}^{(k)} - \mathbf{g}^{(k-1)})}{\alpha^{(k-1)}} = \frac{(\mathbf{g}^{(k)})^T \mathbf{g}^{(k)}}{\alpha^{(k-1)}}.$$

The preceding step-length identity gives

$$(\mathbf{d}^{(k-1)})^T \mathbf{Q} \mathbf{d}^{(k-1)} = \frac{(\mathbf{g}^{(k-1)})^T \mathbf{g}^{(k-1)}}{\alpha^{(k-1)}}.$$

Why the conjugate coefficient simplifies

Start from the adjacent-conjugacy coefficient, with the index shifted:

$$\beta^{(k)} = \frac{(\mathbf{g}^{(k)})^T \mathbf{Q} \mathbf{d}^{(k-1)}}{(\mathbf{d}^{(k-1)})^T \mathbf{Q} \mathbf{d}^{(k-1)}}.$$

The residual recurrence and $(\mathbf{g}^{(k)})^T \mathbf{g}^{(k-1)} = 0$ give

$$(\mathbf{g}^{(k)})^T \mathbf{Q} \mathbf{d}^{(k-1)} = \frac{(\mathbf{g}^{(k)})^T (\mathbf{g}^{(k)} - \mathbf{g}^{(k-1)})}{\alpha^{(k-1)}} = \frac{(\mathbf{g}^{(k)})^T \mathbf{g}^{(k)}}{\alpha^{(k-1)}}.$$

The preceding step-length identity gives

$$(\mathbf{d}^{(k-1)})^T \mathbf{Q} \mathbf{d}^{(k-1)} = \frac{(\mathbf{g}^{(k-1)})^T \mathbf{g}^{(k-1)}}{\alpha^{(k-1)}}.$$

The common factor $1/\alpha^{(k-1)}$ cancels, so

$$\beta^{(k)} = \frac{(\mathbf{g}^{(k)})^T \mathbf{g}^{(k)}}{(\mathbf{g}^{(k-1)})^T \mathbf{g}^{(k-1)}}.$$

Linear CG method

Initialize

$$\mathbf{g}^{(0)} = \mathbf{Q}\mathbf{x}^{(0)} - \mathbf{c}, \quad \mathbf{d}^{(0)} = -\mathbf{g}^{(0)}.$$

For $k = 0, 1, \dots$, while $\|\mathbf{g}^{(k)}\|$ is not sufficiently small:

$$\mathbf{w}^{(k)} = \mathbf{Q}\mathbf{d}^{(k)}, \quad \alpha^{(k)} = \frac{(\mathbf{g}^{(k)})^T \mathbf{g}^{(k)}}{(\mathbf{d}^{(k)})^T \mathbf{w}^{(k)}},$$

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha^{(k)} \mathbf{d}^{(k)}, \quad \mathbf{g}^{(k+1)} = \mathbf{g}^{(k)} + \alpha^{(k)} \mathbf{w}^{(k)},$$

$$\beta^{(k+1)} = \frac{(\mathbf{g}^{(k+1)})^T \mathbf{g}^{(k+1)}}{(\mathbf{g}^{(k)})^T \mathbf{g}^{(k)}}, \quad \mathbf{d}^{(k+1)} = -\mathbf{g}^{(k+1)} + \beta^{(k+1)} \mathbf{d}^{(k)}.$$

Compared with unsimplified $\alpha^{(k)}$ and $\beta^{(k+1)}$, save two inner products:

$$\alpha^{(k)} = -\frac{(\mathbf{g}^{(k)})^T \mathbf{d}^{(k)}}{(\mathbf{d}^{(k)})^T \mathbf{Q}\mathbf{d}^{(k)}}, \quad \beta^{(k+1)} = \frac{(\mathbf{g}^{(k+1)})^T \mathbf{Q}\mathbf{d}^{(k)}}{(\mathbf{d}^{(k)})^T \mathbf{Q}\mathbf{d}^{(k)}}.$$

CG on the common example

The first direction is

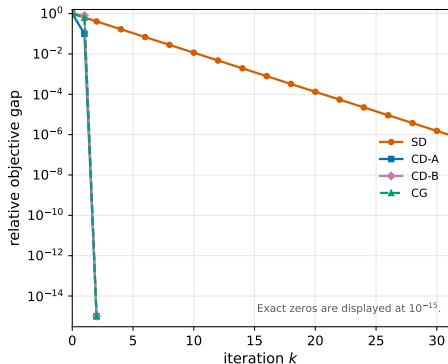
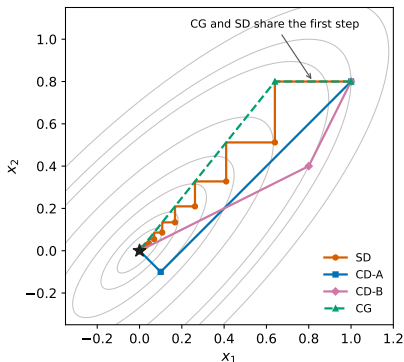
$$\mathbf{g}^{(0)} = (1.8, 0)^T, \quad \mathbf{d}^{(0)} = (-1.8, 0)^T.$$

The two exact-arithmetic updates are

k	$\mathbf{x}^{(k)}$	$q(\mathbf{x}^{(k)})$	$\ \mathbf{g}^{(k)}\ $
0	$(1, 0.8)^T$	0.9	1.8
1	$(0.64, 0.8)^T$	0.576	1.44
2	$(0, 0)^T$	0	0

CG shares its first step with SD. Its second direction retains the first line minimization and reaches the solution.

Trajectories and convergence on one landscape



SD and CG have the same first segment. CD and CG finish in two exact steps, while SD repeatedly crosses the narrow valley.

The classical CG convergence bound

Let $\mathbf{e}^{(k)} = \mathbf{x}^{(k)} - \mathbf{x}^*$ and

$$\kappa(\mathbf{Q}) = \frac{\lambda_{\max}(\mathbf{Q})}{\lambda_{\min}(\mathbf{Q})}.$$

If $\kappa(\mathbf{Q}) = 1$, CG finishes in at most one step. If $\kappa(\mathbf{Q}) > 1$, every generated iterate satisfies

$$\|\mathbf{e}^{(k)}\|_{\mathbf{Q}} \leq 2 \left(\frac{\sqrt{\kappa(\mathbf{Q})} - 1}{\sqrt{\kappa(\mathbf{Q})} + 1} \right)^k \|\mathbf{e}^{(0)}\|_{\mathbf{Q}}.$$

This is a worst-case bound. Exact finite termination and spectral structure can give much faster observed convergence.

Condition-number dependence: SD and CG

For a PD quadratic with $\kappa > 1$ and exact line search, representative worst-case $\mathbf{Q}$ -norm upper bounds are

method	upper bound after k steps
SD	$\left(\frac{\kappa - 1}{\kappa + 1}\right)^k$
CG	$2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}\right)^k + \text{finite termination}$

Preconditioned CG

For

$$q(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} - \mathbf{c}^T \mathbf{x}, \quad \mathbf{Q} \succ 0,$$

choose a fixed $\mathbf{B} \succ 0$ and **change coordinates**:

$$\mathbf{u} = \mathbf{B}^{1/2} \mathbf{x}, \quad \tilde{\mathbf{Q}} = \mathbf{B}^{-1/2} \mathbf{Q} \mathbf{B}^{-1/2}, \quad \tilde{\mathbf{c}} = \mathbf{B}^{-1/2} \mathbf{c}.$$

Then

$$q(\mathbf{x}) = \tilde{q}(\mathbf{u}) = \frac{1}{2} \mathbf{u}^T \tilde{\mathbf{Q}} \mathbf{u} - \tilde{\mathbf{c}}^T \mathbf{u}.$$

method in $\mathbf{u}$ -coordinates	equivalent method in $\mathbf{x}$ -coordinates
SD	preconditioned SD (PSD)
CG	preconditioned CG (PCG)

Preconditioned CG (cont'd)

Write $\mathbf{g}^{(k)} = \mathbf{Q}\mathbf{x}^{(k)} - \mathbf{c}$. Instead of forming $\mathbf{B}^{-1/2}$, solve

$$\mathbf{B}\mathbf{z}^{(k)} = \mathbf{g}^{(k)} \quad (\mathbf{z}^{(k)} = \mathbf{B}^{-1}\mathbf{g}^{(k)}).$$

Initialize $\mathbf{d}^{(0)} = -\mathbf{z}^{(0)}$. Then repeat

$$\mathbf{w}^{(k)} = \mathbf{Q}\mathbf{d}^{(k)},$$

$$\alpha^{(k)} = \frac{(\mathbf{g}^{(k)})^T \mathbf{z}^{(k)}}{(\mathbf{d}^{(k)})^T \mathbf{w}^{(k)}},$$

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha^{(k)} \mathbf{d}^{(k)},$$

$$\mathbf{g}^{(k+1)} = \mathbf{g}^{(k)} + \alpha^{(k)} \mathbf{w}^{(k)},$$

$$\mathbf{B}\mathbf{z}^{(k+1)} = \mathbf{g}^{(k+1)},$$

$$\beta^{(k+1)} = \frac{(\mathbf{g}^{(k+1)})^T \mathbf{z}^{(k+1)}}{(\mathbf{g}^{(k)})^T \mathbf{z}^{(k)}},$$

$$\mathbf{d}^{(k+1)} = -\mathbf{z}^{(k+1)} + \beta^{(k+1)} \mathbf{d}^{(k)}.$$

Effective conditioning and the cost trade-off

Let

$$\kappa_{\text{eff}} := \kappa(\tilde{\mathbf{Q}}), \quad \tilde{\mathbf{Q}} = \mathbf{B}^{-1/2} \mathbf{Q} \mathbf{B}^{-1/2}.$$

PCG inherits the CG bound in the transformed coordinates:

$$\|\mathbf{x}^{(k)} - \mathbf{x}^*\|_{\mathbf{Q}} \leq 2 \left(\frac{\sqrt{\kappa_{\text{eff}}} - 1}{\sqrt{\kappa_{\text{eff}}} + 1} \right)^k \|\mathbf{x}^{(0)} - \mathbf{x}^*\|_{\mathbf{Q}}.$$

- $\mathbf{B} = \mathbf{I}$ recovers CG. With $\mathbf{B} = \mathbf{Q}$, PCG takes one step, but solving with $\mathbf{B}$ is then the original problem.
- Common choices include [diagonal or block-diagonal matrices](#) and preconditioners built from [incomplete Cholesky factors](#).
- A useful $\mathbf{B}$ [lowers \$\kappa_{\text{eff}}\$ enough to repay the extra solve with \$\mathbf{B}\$](#) .

A Jacobi-PCG example

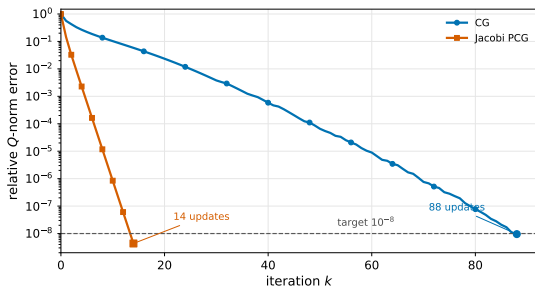
Let $n = 100$ and $\mathbf{Q} = \mathbf{D}\mathbf{T}\mathbf{D}$, where

$$\mathbf{T} = \text{tridiag}\left(-\frac{1}{4}, 1, -\frac{1}{4}\right), \quad (\mathbf{D})_{ii} = 10^{(i-1)/(n-1)}.$$

Set $\mathbf{x}^* = \mathbf{1}$, $\mathbf{c} = \mathbf{Q}\mathbf{1}$, and $\mathbf{x}^{(0)} = \mathbf{0}$. For the Jacobi preconditioner,

$$\mathbf{B} = \text{diag}(\mathbf{Q}) = \mathbf{D}^2, \quad \tilde{\mathbf{Q}} = \mathbf{B}^{-1/2}\mathbf{Q}\mathbf{B}^{-1/2} = \mathbf{T}.$$

method	κ	updates
CG	220.94	88
PCG	2.998	14



From linear to nonlinear CG

For a general smooth function, set $\mathbf{g}^{(k)} = \nabla f(\mathbf{x}^{(k)})$ and retain the short recurrence

$$\mathbf{d}^{(0)} = -\mathbf{g}^{(0)}, \quad \mathbf{d}^{(k)} = -\mathbf{g}^{(k)} + \beta^{(k)}\mathbf{d}^{(k-1)},$$

choose $\alpha^{(k)} > 0$ by a line search, and update

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha^{(k)}\mathbf{d}^{(k)}.$$

- (Usually) no exact conjugacy.
- (Usually) no descent-direction guarantee.
- (Usually) no exact line search.

Nonlinear conjugate-gradient coefficients

Let

$$\mathbf{y}^{(k-1)} = \mathbf{g}^{(k)} - \mathbf{g}^{(k-1)}.$$

Common choices are

name	$\beta^{(k)}$
Fletcher–Reeves (FR), 1964	$\frac{(\mathbf{g}^{(k)})^T \mathbf{g}^{(k)}}{(\mathbf{g}^{(k-1)})^T \mathbf{g}^{(k-1)}}$
Polak–Ribière–Polyak (PRP), 1969	$\frac{(\mathbf{g}^{(k)})^T \mathbf{y}^{(k-1)}}{(\mathbf{g}^{(k-1)})^T \mathbf{g}^{(k-1)}}$
Hestenes–Stiefel (HS), 1952	$\frac{(\mathbf{g}^{(k)})^T \mathbf{y}^{(k-1)}}{(\mathbf{d}^{(k-1)})^T \mathbf{y}^{(k-1)}}$
Dai–Yuan (DY), 1999	$\frac{(\mathbf{g}^{(k)})^T \mathbf{g}^{(k)}}{(\mathbf{d}^{(k-1)})^T \mathbf{y}^{(k-1)}}$

For a strictly convex quadratic with exact line search, **they agree with linear CG in exact arithmetic.**

Safeguards for nonlinear CG

- **Inexact line-search conditions** replace an impractical exact line search and support convergence analysis.

- **PRP⁺** uses

$$\beta_{\text{PRP}^+}^{(k)} = \max\{0, \beta_{\text{PRP}}^{(k)}\}.$$

It often performs robustly in practice, but no variant is uniformly best.

- If $(\mathbf{g}^{(k)})^T \mathbf{d}^{(k)}$ fails a chosen descent safeguard, **restart** with $\mathbf{d}^{(k)} = -\mathbf{g}^{(k)}$.
- **Periodic or orthogonality-based restarts** limit accumulated loss of quadratic structure.

Take-home messages

- CD: memory over an expanding subspace.
- CG: recursive and cheap constructions from $\mathbf{g}$ and $\mathbf{Qd}$.
- CG convergence rate: dependence on $\sqrt{\kappa}$, PCG.
- Nonlinear CG: $\beta^{(k)}$, inexact line-search, descent safeguards, restarts.

Reference: Krylov subspaces generated by the recurrence

Define

$$\mathcal{K}_{k+1}(\mathbf{Q}, \mathbf{g}^{(0)}) := \text{span}\{\mathbf{g}^{(0)}, \mathbf{Q}\mathbf{g}^{(0)}, \dots, \mathbf{Q}^k \mathbf{g}^{(0)}\}.$$

Until early termination, the same recursive properties imply

$$\text{span}\{\mathbf{d}^{(0)}, \dots, \mathbf{d}^{(k)}\} = \text{span}\{\mathbf{g}^{(0)}, \dots, \mathbf{g}^{(k)}\} = \mathcal{K}_{k+1}(\mathbf{Q}, \mathbf{g}^{(0)}).$$

Thus one initial gradient and repeated multiplication by $\mathbf{Q}$ generate the entire correction space. The span identity, orthogonality, and conjugacy belong to the same joint induction.

Reference proof: residual orthogonality

For $i < k$, expand the gradient recurrence from iteration $i + 1$:

$$\mathbf{g}^{(k)} = \mathbf{g}^{(i+1)} + \sum_{j=i+1}^{k-1} \alpha^{(j)} \mathbf{Q} \mathbf{d}^{(j)}.$$

The exact search at iteration i gives $(\mathbf{g}^{(i+1)})^T \mathbf{d}^{(i)} = 0$. Pairwise conjugacy gives $(\mathbf{d}^{(i)})^T \mathbf{Q} \mathbf{d}^{(j)} = 0$ for $j > i$. Therefore

$$(\mathbf{g}^{(k)})^T \mathbf{d}^{(i)} = 0, \quad i = 0, \dots, k-1.$$

Since q is strictly convex, these first-order conditions show that $\mathbf{x}^{(k)}$ uniquely minimizes q over $\mathbf{x}^{(0)} + \text{span}\{\mathbf{d}^{(0)}, \dots, \mathbf{d}^{(k-1)}\}$.

Reference proof: finite termination

Suppose

$$\sum_{i=0}^m \gamma_i \mathbf{d}^{(i)} = \mathbf{0}.$$

Multiplying by $(\mathbf{d}^{(j)})^T \mathbf{Q}$ gives

$$\gamma_j (\mathbf{d}^{(j)})^T \mathbf{Q} \mathbf{d}^{(j)} = 0.$$

Positive definiteness and $\mathbf{d}^{(j)} \neq \mathbf{0}$ imply $\gamma_j = 0$ for every j . Thus nonzero conjugate directions are linearly independent. After n directions,

$$\text{span}\{\mathbf{d}^{(0)}, \dots, \mathbf{d}^{(n-1)}\} = \mathbf{R}^n.$$

Expanding-subspace optimality then gives $\mathbf{x}^{(n)} = \arg \min_{\mathbf{x} \in \mathbf{R}^n} q(\mathbf{x}) = \mathbf{x}^*$.

Reference proof: CG conjugacy

Before termination, assume conjugacy through $\mathbf{d}^{(k)}$. The adjacent coefficient makes $(\mathbf{d}^{(k+1)})^T \mathbf{Q} \mathbf{d}^{(k)} = 0$. The triangular recurrence and gradient update give

$$\text{span}\{\mathbf{g}^{(0)}, \dots, \mathbf{g}^{(j)}\} = \text{span}\{\mathbf{d}^{(0)}, \dots, \mathbf{d}^{(j)}\} = \mathcal{K}_{j+1}(\mathbf{Q}, \mathbf{g}^{(0)}),$$

and

$$\mathbf{Q} \mathbf{d}^{(i)} = \frac{\mathbf{g}^{(i+1)} - \mathbf{g}^{(i)}}{\alpha^{(i)}} \in \text{span}\{\mathbf{d}^{(0)}, \dots, \mathbf{d}^{(i+1)}\}.$$

For $i < k$, exact-search orthogonality therefore yields $(\mathbf{g}^{(k+1)})^T \mathbf{Q} \mathbf{d}^{(i)} = 0$. Hence

$$(\mathbf{d}^{(k+1)})^T \mathbf{Q} \mathbf{d}^{(i)} = -(\mathbf{g}^{(k+1)})^T \mathbf{Q} \mathbf{d}^{(i)} + \beta^{(k+1)} (\mathbf{d}^{(k)})^T \mathbf{Q} \mathbf{d}^{(i)} = 0.$$

Induction proves pairwise conjugacy.

Reference proof idea: the CG rate

Krylov optimality gives the polynomial bound

$$\|\mathbf{e}^{(k)}\|_{\mathbf{Q}} \leq \min_{p \in \mathcal{P}_k, p(0)=1} \|p(\mathbf{Q})\mathbf{e}^{(0)}\|_{\mathbf{Q}} \leq \min_{p \in \mathcal{P}_k, p(0)=1} \max_{\lambda \in [\lambda_{\min}, \lambda_{\max}]} |p(\lambda)| \|\mathbf{e}^{(0)}\|_{\mathbf{Q}}.$$

Map $[\lambda_{\min}, \lambda_{\max}]$ to $[-1, 1]$ and choose the scaled Chebyshev polynomial with value one at $\lambda = 0$. Its extremal property gives

$$\max_{\lambda \in [\lambda_{\min}, \lambda_{\max}]} |p(\lambda)| \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k.$$

Substitution yields the classical bound. The lecture notes give the complete polynomial argument.